

Cristóbal Alcázar

MACHINE LEARNING · EXPERIMENTATION · GENERATIVE AI

Santiago, Chile

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Work Experience

Thoughtworks - LATAM Airlines (eBusiness)

Santiago, Chile

DATA SCIENTIST

Dec. 2024 - Today

- Designed and implemented a unified ML serving and orchestration platform (Python, GCP) with a probabilistic decision layer (select, sample, rank) for personalized product recommendations, decoupling model integration from the front-end – adopted as standard by two independent teams; contributed Kubeflow/Vertex AI Pipelines components to the org's internal MLOps framework, now used across multiple model repos and adopted by at least one other product.
- Grew the app's experimentation program from a single live test to a 30+ experiment portfolio, taking the app from 0 to top 3 within eBusiness; cross-sell initiatives applying A/B testing and causal inference captured USD \$18MM in 2025.
- Identified an experiment-brief validation gap in a cross-team workshop and proposed brief-reviewer, an LLM-based tool now embedded in LATAM's experimentation agent (Cosmos); scoped the implementation and contributed directly to the agent, prioritized onto the MLOps workflow roadmap and live in production today.

Cofounder - Vendie

Santiago, Chile

AI-POWERED WHATSAPP COMMERCE PLATFORM FOR SMALL RETAILERS

Oct. 2023 - 2026

- Won **Open Salcobrand 2024** to pilot conversational commerce in a single store, then scaled to full Santiago coverage; landed small-merchant clients – the target segment – but wound down in 2026 as onboarding friction (many couldn't access their own Facebook Business accounts) strained operations faster than new customer acquisition could offset.
- Designed and built a provider-agnostic agentic framework from scratch – LLM/agent abstractions, pydantic-schema function calling across multiple model providers, automatic fallback – orchestrated by a router agent delegating to specialized agents (semantic product search over a GCP vector database, address capture, FAQ, order flow) on a serverless architecture with full observability. [\[Slides\]](#)

European Southern Observatory - Paranal Observatory

Paranal, Chile

MACHINE LEARNING ENGINEER (GRADUATE INTERNSHIP) - SOFTWARE GROUP

Aug. 2023 - Jan. 2024

- Built a semantic-search engine over ESO's on-premise Jira ticket history – issue extractor, chunker, embedding pipeline, and OpenSearch retriever – letting engineers query an instrument fault (e.g. GRAVITY) and surface the most similar past resolutions; first AI-powered search layer on a system with no vendor AI features available.
- Built an LLM-based shift-handoff reporting workflow – general summary, highlights tagged by operational nomenclature codes, and data-source references – fed by a Jira/Subversion aggregator API and editable by the engineer before sending, turning report drafting into a collaborative tool rather than rigid AI output. [\[Slides\]](#)

Compraquí - BancoEstado subsidiary

Santiago, Chile

PAYOPS AND RISK/FRAUD LEAD

Oct. 2019 - Oct. 2020

- Owned payment-processing operations – the core transaction lifeblood of a payments-aggregation business – for a merchant portfolio that scaled from 8k to 50k+ nationwide, building the fraud and risk layer that safeguarded transaction quality through the joint venture's growth phase, ending with the portfolio's sale to SumUp.
- Built fraud-detection features and prototyped an anomaly-detection model over transaction amount, merchant-category deviation, and frequency, used to block or reactivate risky merchants.

MANAGER, PAYOPS AND ANALYTICS

Oct. 2020 - May. 2021

- Identified data access as a strategic blocker – the new processor's default reporting was Excel/CSV-only – and led the initial buildout of an ETL layer on AWS to make transaction data queryable, the foundation needed to mature fraud detection beyond ad hoc analysis; adapted the risk-monitoring dashboard (R) – used by risk analysts to flag and act on suggested risks – to the new platform, alongside weekly executive analytics that became standard reporting for managers.
- Co-designed core acquiring processes for BancoEstado's first-ever acquiring operation – historically Transbank's alone – with Compraquí as its only pilot client: reconciliation workflows, MCC-based risk thresholds letting pre-onboarded small merchants transact within limits instead of blocking them outright, and chargeback/reversal SLAs, all built from scratch.

Central Bank of Chile

Santiago, Chile

ECONOMIC FINANCIAL ANALYST - MACROECONOMIC STATISTIC DIVISION

Oct. 2016 - Oct. 2019

- Built statistics from scratch where no structured data existed – an R package normalizing unstructured Excel/PDF financial statements into a company balance-sheet registry, and Python scrapers classifying shareholders and bondholders by economic sector via custom classification dictionaries and imputation models.
- Owned Chile's national debt-statistics pipeline (SQL/SAS), implementing the IMF's methodology for Chile's accounting reality; still in production today as a core input to Chile's quarterly national institutional accounts, and the base for the publicly released *Handbook of Securities Statistics* (SDDS Plus).

Research & Teaching Experience

RESEARCH

- 2025 **Using Language Models to Detect and Reduce Gender Bias in University Forum Messages**, Designed the debiasing agent workflow. Co-authored with G. Salomó-López, R. Barceló, C. Reyes, D. Radovic & F. Tobar, *IEEE Signal Processing Magazine* [\[Paper\]](#) [\[GitHub\]](#) *Santiago, Chile*
- 2024 **Avoiding mode collapse on diffusion models finetuned with reinforcement learning**, Research Assistant worked with [@Felipe Tobar](#) and [@Roberto Barceló](#) [\[arXiv\]](#) *Santiago, Chile*
- 2022 **SegPlantFormer**, An automatic cell plants segmentation from high-resolution microscope images using transformer-based approach. Collaboration with researchers of Worcester Polytechnic Institute (US) [\[GitHub\]](#) [\[Poster\]](#) *Santiago, Chile*

TEACHING EXPERIENCE & TALKS

- 2024 **European Southern Observatory**, Building LLM-based Systems & Products at Paranal [\[Slides\]](#) *Santiago, Chile*
- 2023 **Universidad de Chile - FCFM**, Graduate course: Deep Generative Models (Teaching Assistant - Syllabus design) *Santiago, Chile*
- 2023 **European Southern Observatory**, Talk: Sequence Models with Transformers. A Journey Through Language Models, Current Insights, and Practical Applications at Paranal Observatory (with [@Camilo Carvajal](#)). [\[Slides\]](#) *Paranal, Chile*
- 2022/23 **Universidad Tecnológica Metropolitana**, Diploma course: Modelamiento de datos (Teaching Assistant - Syllabus design) *Santiago, Chile*
- 2021 **Women in Data - Guatemala Chapter (NPO)**, Data Visualization with Python [\[Slides\]](#) [\[GitHub\]](#) [\[Recording\]](#) *Webinar - Remote*
- 2019 **Universidad de Chile - FCFM**, Undergraduate course: Data Journalism with Python (Part-time Professor) *Santiago, Chile*
- 2019 **Ministry of Social Development and Family**, A five-session Data Visualization Workshop using R (Instructor - Syllabus design) *Santiago, Chile*
- 2018 **Universidad de Chile - FCEI**, Undergraduate course: Data Analysis with R Programming (Part-time Professor) *Santiago, Chile*
- 2018 **Universidad de Chile - FCFM**, Undergraduate course: Data Journalism with R (Teaching Assistant) *Santiago, Chile*

Education

Universidad de Chile

MSC IN DATA SCIENCE

Santiago, Chile

2022 - 2024

- **Thesis:** *Extending Reinforcement Learning Techniques for Diffusion Models*, supervised by [@Felipe Tobar](#) [\[Report\]](#) [\[Slides\]](#) [\[GitHub\]](#)

Universidad Adolfo Ibáñez

B.S. IN BUSINESS ADMINISTRATION, MSC IN FINANCE

Viña del Mar, Chile

2009 - 2015

Writing

How I Powered Up My Deep Learning Workflow with HuggingFace and W&B

Weights & Biases (Official Blog)

AUTHOR

Feb. 2023

- Featured community post; companion piece to the SegPlantFormer computer vision research project, covering experiment tracking and reproducible training pipeline design [\[Post\]](#).

Digital literacy, computational thinking, and literate programming

Lom Ediciones

AUTHOR

Oct. 2019

- A chapter published in the book *"Inteligencia Artificial y Bienestar de las juventudes en America Latina"* ([Brossi, 2019](#)) that reviews and reflects on ideas such as computational thinking and literate programming, particularly highlighting the importance of establishing programming as an elementary skill in Latin America.